



Security Threats

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Why Cybersecurity Matters ?

Colonial Oil Pipeline, 2021

- ▶ Colonial Oil Pipeline is one of the largest pipelines in the US, supplies 45% of the East Coast fuel.
- ▶ On May 2021 the Colonial Oil Pipeline Company was attacked by ransomware.
- ▶ <https://www.techtarget.com/whatis/feature/Colonial-Pipeline-hack-explained-Everything-you-need-to-know>
- ▶ Network and operations completely shut down for some days.
 - ▶ No Operational Technology that moves oil compromised.
 - ▶ IT network and computer systems attacked by ransomware.
- ▶ Social impact: “Fear of a gas shortage caused panic-buying and long lines at gas stations in many states”.

Mirai Botnet, 2016

- ▶ During 2016 multiple Distributed Denial of Service (DDOS) attacks appear:
 - ▶ OVH, a france telecom, was hit by ddos of unprecendent scale (1.1 terabits per second)
 - ▶ Dyn, us telecom part of Internet backbone, slowed or stopped the internet for eastern coast.
- ▶ <https://www.wired.com/story/mirai-botnet-minecraft-scam-brought-down-the-internet/>
- ▶ The Mirai botnet:
 - ▶ was capable of scanning the internet for different IOT devices,
 - ▶ unsecured internet-of-things devices like security cameras and wireless routers,
 - ▶ with manufacturers' default security settings;
 - ▶ command and control more than 600,000 devices around the world.

Common Terms

Assets, Vulnerability, Threat, Exploit, Mitigation, Risk

Asset

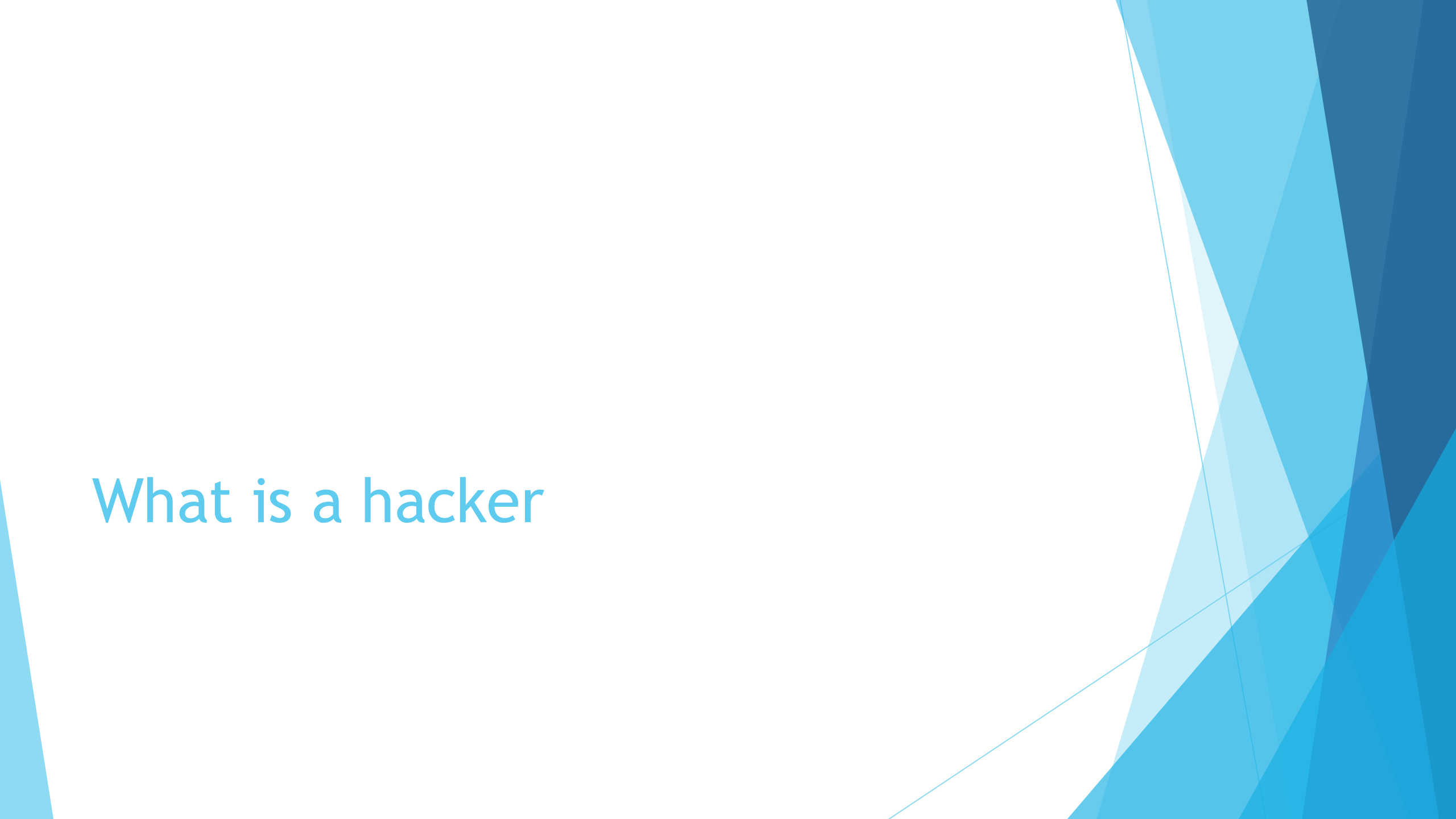
- ▶ **Assets** are all items of value to the users of a system.
- ▶ Examples of assets:
 - ▶ hardware (physical components of the computer and networks),
 - ▶ software (programs, operating systems)
 - ▶ data (or information) (specific to users).
- ▶ The value of assets depends on their utility, perspective and time.
- ▶ Because of the value of assets, they are targets for attack.

Vulnerability, Threat, Attack

- ▶ A **vulnerability** is a weakness on an asset that can be exploited to cause harm or damage.
- ▶ A **threat** is a set of circumstances that can potentially cause damage to the asset.
- ▶ An **attack** is the realization of a threat that exploits a vulnerability to cause damage to the asset.

Risk Management

- ▶ Risk is the expectation of loss expressed as the probability of getting a particular harmful result from the exploitation of a threat.
- ▶ Risk Management deals with mitigating or reducing the impact of risks on the system.
- ▶ Countermeasures are actions that protect the system (its assets) in order to reduce its vulnerability and the risk overall.

The background features a series of overlapping, semi-transparent blue triangles and polygons of various shades, ranging from light sky blue to deep navy blue. These shapes are primarily concentrated on the right side of the frame, creating a dynamic, modern geometric pattern. The left side of the image is a solid, very light blue, providing a clean backdrop for the text.

What is a hacker

What is a Hacker

- ▶ A Hacker is someone who applies their computer skills toward solving a problem.
 - ▶ No definition on the intention or legality of the action.
 - ▶ <https://www.avast.com/c-hacker>
- ▶ Black Hat hacker: technical hacker that uses its skills in unlawful access to computer and network systems, usually for profit, fame or to create havoc.
 - ▶ Typical actions are to gain (unauthorised) access to systems;
 - ▶ and cause as consequences the disrupting normal operations,
 - ▶ or gaining access to private data in order to benefit from it by deleting it, stealing and selling it.
- ▶ In the previous examples:
 - ▶ The colonial oil pipeline was a case of ransomware (to obtain money in exchange of normal operation, that resulted in locked or shut down systems);
 - ▶ The Mirai botnet was a case of disrupting normal operations of competing servers, through the use of denial of service attacks.

What is a Hacker

- ▶ White Hat hacker: technical hacker that uses its skills for securing systems.
 - ▶ Also known as *ethical hacking*.
 - ▶ The skills and tools used are often the same between white and black hats.
 - ▶ Whoever, in this case actions are consent by the owners, and vulnerabilities are documented to the owners before public dissemination.
- ▶ Grey Hat hacker: a hacker that sits between pure ethical hacking and criminal hacking activities:
 - ▶ Although not initially motivated by profit or ill-intended, the actions are made in a rush without consent, causing some consequences.
 - ▶ <https://edition.cnn.com/2013/08/19/tech/social-media/zuckerberg-facebook-hack>

Common Network Attacks

Common Network Attacks (by Tripwire)

- ▶ Reconnaissance attacks
 - ▶ *Reconnaissance attacks are centered around general knowledge gathering.*
 - ▶ *Some common examples of reconnaissance attacks include packet sniffing, ping sweeps, port scanning, phishing, social engineering, and internet information queries.*
- ▶ Access attacks
 - ▶ *Unauthorized access refers to attackers accessing a network without receiving permission.*
 - ▶ *Among the causes of unauthorized access attacks are weak passwords, lacking protection against social engineering, previously compromised accounts, and insider threats.*
- ▶ (Distributed) DoS attacks
 - ▶ *Attackers build botnets, large fleets of compromised devices, and use them to direct false traffic at your network or servers.*
 - ▶ *DDoS can occur at the network level, for example by sending huge volumes of SYN/ACC packets which can overwhelm a server.*

<https://www.tripwire.com/state-of-security/3-types-of-network-attacks>

Common Network Attacks (by Fortinet)

▶ Denial-Of-Service

- ▶ *A DoS attack is designed to overwhelm the resources of a system to the point where it is unable to reply to legitimate service requests.*
- ▶ *A DDoS attack is initiated by a vast array of malware-infected host machines controlled by the attacker.*

▶ Man In The Middle (MITM)

- ▶ *MITM attacks make it possible for an attacker to eavesdrop on the data sent back and forth between two parties.*
- ▶ *The two parties involved feel like they are communicating as they normally do. What they do not know is that the person actually sending the message illicitly modifies or accesses the message before it reaches its destination.*

▶ DNS spoofing

- ▶ *A hacker alters DNS records to send traffic to a fake or “spoofed” website. Once on the fraudulent site, the victim may enter sensitive information that can be used or sold by the hacker.*
- ▶ *In a DNS spoofing attack, the attacker takes advantage of the fact that the user thinks the site they are visiting is legitimate.*

<https://www.fortinet.com/resources/cyberglossary/types-of-cyber-attacks>

Common Network Attacks (by Fortinet)

- ▶ Phishing
 - ▶ *A phishing attack occurs when a malicious actor sends emails that seem to be coming from trusted, legitimate sources in an attempt to grab sensitive information from the target.*
- ▶ Password Attack
 - ▶ A Password attack enables an intruder to pretend to be a legitimate user when authenticating into a system.
 - ▶ Typical attacks are brute-force attack and dictionary attack.
- ▶ SQL injection
 - ▶ *an SQL attack inserts or “injects” an SQL command into a database, in place of some data, such as a password or login. The server that holds the database then runs the command and the system is penetrated.*
 - ▶ *If an SQL injection succeeds, several things can happen, including the release of sensitive data or the modification or deletion of important data.*

<https://www.fortinet.com/resources/cyberglossary/types-of-cyber-attacks>

Common Network Attacks (by Fortinet)

▶ Ransomware

- ▶ *With Ransomware, the victim's system is held hostage until they agree to pay a ransom to the attacker.*
- ▶ *After the payment has been sent, the attacker then provides instructions regarding how the target can regain control of their computer.*
- ▶ *Sometimes, even after paying the ransom, it is not technically possible to recover the system.*

▶ Malware

- ▶ *Malware is a general term for malicious software. Malware infects a computer and changes how it functions, destroys data, or spies on the user or network traffic as it passes through.*
- ▶ *In a malware attack, the software has to be installed on the target device.*

<https://www.fortinet.com/resources/cyberglossary/types-of-cyber-attacks>